

Wireless Security



B. Tech CSE, VI Semester

Instructor:

Dr Mohammad Wazid

Professor, Department of Computer Science and Engineering

Head of Cybersecurity and Internet of Things Research Group

Graphic Era (Deemed to be University), Dehradun, India

Wireless Security: Topics

- Overview of wireless security
- Threats related to wireless security
- Wireless security protocols

Wireless LAN

- A Wireless Local Area Network (WLAN), or wireless LAN, is a type of local area network (LAN) that uses wireless communication, like Wi-Fi, to connect devices instead of physical cables.
- A WLAN is a network that allows devices to connect and communicate wirelessly within a limited area, such as a home, office, or school.
- WLANs use radio waves to transmit data between devices, like laptops, smartphones, and tablets, and a central access point (often a router).

Wireless LAN

- WLANs provide flexibility and mobility, allowing users to move around while staying connected to the network.
- Typically operate within a specific geographic area, like a building or campus.
- Most WLANs are based on the IEEE 802.11 standards, commonly known as Wi-Fi.
- WLANs use access points (APs) to create a wireless network and allow devices to connect.

Scenario of a wireless LAN

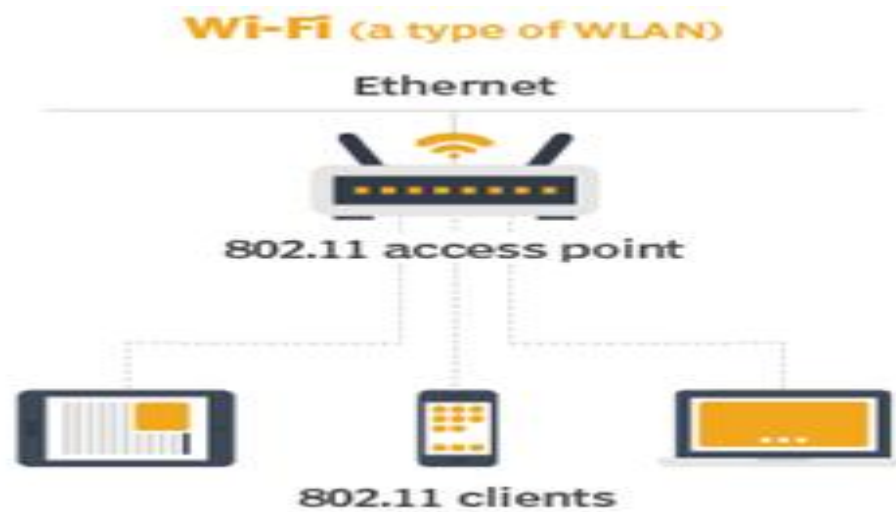


Fig. Scenario of a wireless LAN

Overview of wireless security

- Wireless security refers to measures used to protect wireless networks from **unauthorized access, attacks, and data breaches.**
- Unlike wired networks, wireless networks transmit data **over radio waves, making them more vulnerable to interception and attacks.**

Overview of wireless security

- **Problems with wireless communication:**
- **Open transmission medium** (susceptible to eavesdropping).
- **Weak encryption** in older security protocols.
- **Risk of rogue access points and Man-in-the-Middle (MITM) attacks.**
- **Unauthorized access and denial-of-service (DoS) attacks.**

Overview of wireless security

- **General wireless technologies:**
- **Wi-Fi (802.11 Standards)-** Used for LANs and public hotspots.
- **Bluetooth-** Short-range, personal area networks (PANs).
- **WiMAX (802.16)-** Wireless broadband networks.

Threats to wireless security

- Rogue Access Points
- Evil Twin Attack
- Man-in-the-Middle (MITM)
- Deauthentication Attacks
- Wireless Sniffing

Threats to wireless security

- **1. Rogue access points (APs)**
- Attackers set up unauthorized APs that mimic legitimate networks to intercept user data.
- **Solutions to this attack:**
- **Enable MAC filtering.**
- Use Wireless Intrusion Detection Systems (**WIDS**).

Threats to wireless security

- **2. Evil twin attack**

- Similar to a rogue AP, where an attacker creates a duplicate SSID (Service Set Identifier) to trick users into connecting.

Solutions to this attack.

- **Verify SSID and certificate** before connecting.
- **Use VPNs to encrypt traffic.**

Threats to wireless security

- **3. Man-in-the-Middle (MITM) Attacks**
- An attacker intercepts communication between a user and a legitimate server.
- **Solutions to this attack.**
- Use HTTPS and VPN encryption.
- Disable open Wi-Fi networks.

Threats to wireless security

- **4. Deauthentication (Deauth) attacks**
- Attackers send fake deauthentication packets to disconnect users and force reconnections.
- **Solutions to this attack.**
- Implement 802.11w Protected Management Frames (PMF)

Threats to wireless security

- **5. Wireless sniffing and eavesdropping**
- Attackers use tools like Wireshark or Kismet to capture unencrypted wireless traffic.
- **Solutions to this attack.**
- **Use WPA3 encryption.**
- **Disable SSID broadcasting.**

Wireless security protocols

- **1. WEP (Wired Equivalent Privacy) [Deprecated]**
- Encryption: RC4 (Weak)
- Key Length: 40-bit / 104-bit
- **Vulnerabilities:**
- **Easily cracked using packet injection (Aircrack-ng).**
- No protection against **replay attacks.**
- **Static encryption keys (no session-based security).**

Wireless security protocols

- **2. WPA (Wi-Fi Protected Access) [Legacy Protocol]**
- **Encryption: TKIP (Temporal Key Integrity Protocol)**
- Key Length: 128-bit
- **Improvements Over WEP:**
- **Dynamic key generation (per-packet key mixing)**
- **Message Integrity Check (MIC)** to prevent packet modification
- Still **vulnerable to brute force and dictionary attacks**

Wireless security protocols

- **3. WPA2 (Wi-Fi Protected Access 2) [Industry Standard]**
- Encryption: **AES-CCMP (Advanced Encryption Standard - Counter Mode Cipher Block Chaining Message Authentication Code Protocol)**
- Key Length: 256-bit
- **Security Features:**
- **Strong encryption** using AES
- **Protection against replay attacks**
- **Support for 802.1X authentication** (Enterprise mode)
- **Vulnerabilities: WPA2-PSK susceptible to brute-force attacks and key reinstallation attack (i.e., guessing of key).**

Wireless security protocols

- **4. WPA3 [Modern Wireless Security Standard]**
- Encryption: **AES-256-GCMP (Galois/Counter Mode Protocol)**
- Key Exchange: **Simultaneous Authentication of Equals (SAE)**
- **Security Features:**
- **Prevents dictionary attacks** using forward secrecy.
- **Individualized Data Encryption** for public Wi-Fi security
- 192-bit encryption for enterprise networks
- **Problems: Compatibility issues with older devices.**

Wireless security mechanisms

- **1. Authentication methods**
- **Pre-Shared Key (PSK)-** Used in home networks, vulnerable to brute force attacks.
- **802.1X Authentication-** Used in enterprise environments with a RADIUS server.
- **Multi-Factor Authentication (MFA)-** Requires additional verification beyond passwords.

A RADIUS (Remote Authentication Dial-In User Service) server is a central server that handles authentication, and authorization for wireless networks.

Wireless security mechanisms

- **2. Encryption techniques**
- **AES (Advanced Encryption Standard)**- Used in WPA2 and WPA3, strongest encryption.
- **TKIP (Temporal Key Integrity Protocol)**- Used in WPA, now deprecate.
- **RC4 (Rivest Cipher 4)**- Used in WEP, weak and easily cracked.

3. Wireless Intrusion Detection Systems (WIDS):

Monitors wireless traffic for unauthorized access points, MAC spoofing, and deauthentication attacks.

References

- Cryptography and Network Security: Principles and Practice, text book by William Stallings.